IoT Homework 2025/2026

PART 1– 802.15.4

Exercise 1 (1/3)

Consider the following pseudocode for an ESP32-based vibration monitoring node.

```
// Global timer handle
declare timer_handle as esp_timer_handle_t

function setup_accelerometer():
    initialize_accelerometer(sampling_rate = 100 Hz)

function setup_timer():
    create_periodic_timer(callback = process_window,
                          period = 10 seconds)

function process_window(arg):
    vibration_peaks = count_vibration_peaks(last_10_seconds)

    if vibration_peaks == 0:
        payload = create_message(size = 512 bytes)
    else if vibration_peaks == 1:
        payload = create_message(size = 2 KB)
    else:
        payload = create_message(size = 4 KB)

    send_payload(payload)

function app_main():
    setup_accelerometer()
    setup_timer()

    loop forever:
        delay(100 ms)
```

Exercise 1 (2/3)

Assume that the number of vibration peaks detected in each 10-second window follows a Poisson distribution with average rate: λ=0.4 peaks/window.

The system is operated with IEEE 802.15.4 in beacon-enabled mode, CFP only.
Assume: R = $250kbps$, L = $128bytes$
where $\boldsymbol{R}$ is the nominal bit rate and $\boldsymbol{L}$ is the packet size.

Assume that:

- one packet fits exactly in one CFP slot;
- the CFP is dimensioned according to the expected traffic load;
- the reporting period is 10 seconds;
- the monitoring system is composed of 1 PAN coordinator and 3 vibration monitoring nodes

Exercise 1 (3/3)

Questions:

1. Compute the Probability Mass Function of the output rate: $\boldsymbol{P(r=r0), P(r=r1), P(r=r2)}$ where $r0$, $r1$, and $r2$ are the output rates when the node detects 0, 1, or more than 1 vibration peaks, respectively.
2. Based on the output rate PMF, compute a consistent CFP slot assignment for the system.
3. Compute:
    - Ts
    - $Nslots$ in the CFP
    - $Tactive$
    - $Tinactive$
    - The duty cycle of the system
4. How many additional vibration monitoring nodes can be added while keeping the duty cycle below 10%?

Answer in PDF **Exercise1.pdf**.

PART 2 – BLE Localization

Exercise 2 (1/2)

A nursing home uses an IoT system to help caregivers localize dental prostheses of Alzheimer patients.
Each denture contains a small BLE beacon that periodically transmits advertising packets. BLE gateways installed in the nursing home receive the packets and estimate the approximate position of the denture based on RSSI measurements.

The current firmware is the following:

```
loop forever:
    send_BLE_beacon()
    sleep(10 seconds)
```

The objective is to improve the firmware and the sensing architecture in order to increase battery lifetime while preserving the localization service.

Exercise 2 (2/2)

Design an improved IoT solution.

Your answer should include:

1. The logic used to for BLE beaconing;
2. Any additional sensors added to the system;
3. Pseudocode of the improved firmware;
4. One or more real commercial components that could be used to implement the proposed sensing logic;
5. An estimate of the expected energy saving with respect to the original periodic beaconing solution.

In the energy-saving estimate, clearly state all assumptions including the average time per day in which the denture is expected to require localization

Answer in file **Exercise2.pdf**

PART 3 – RFID

Exercise 3 - RFID

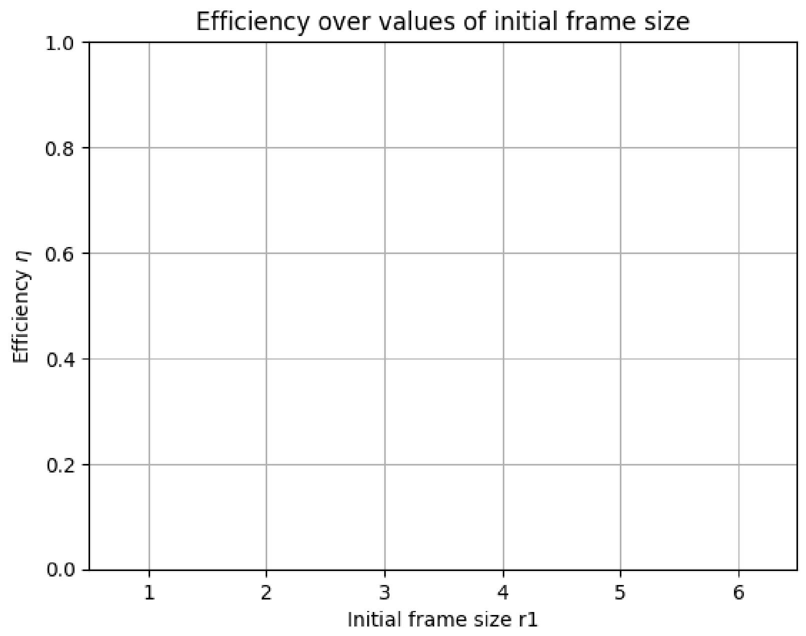
A RFID system based on Dynamic Frame ALOHA is composed of N=4 tags

1. Find the overall collision resolution efficiency **η** in the different cases in which the initial frame size is set to **r1=**1,2,3,4,5,6

- Assume that after the first frame, the frame size is correctly set to the current backlog size
- Assume as given the duration of the arbitration period with N=2,3 tags when r=N

  (**L_2=4, L_3=51/8**)

2. After computing the values of the efficiency with the different frame sizes, **produce a plot** with values of **η over r1** (as figure) -> add in the report



3. For what values of **r1** we have the maximum value for **η? Briefly comment this.**

Answer in PDF Exercise3.pdf